

Lecture Nineteen

AR models

$$y_t = \phi_0 + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

AR(p)

$$\varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$$

y_1
 y_2
 $\vdots$
 y_p

OLS
AutoReg

$$\hat{\phi}_j, \hat{\sigma}^2$$

$$\begin{aligned} \hat{y}_{n+1} &= \hat{\phi}_0 + \hat{\phi}_1 y_n + \hat{\phi}_2 y_{n-1} + \dots + \hat{\phi}_p y_{n+1-p} \\ \hat{y}_{n+2} &= \hat{\phi}_0 + \hat{\phi}_1 \hat{y}_{n+1} + \hat{\phi}_2 y_n + \dots + \hat{\phi}_p y_{n+2-p} \\ \hat{y}_{n+3}, \dots \end{aligned}$$

get-prediction

$$\text{Cov} \left(\begin{array}{c} y_{n+1} \\ \vdots \\ y_{n+k} \end{array} \middle| \begin{array}{c} y_1, \dots, y_n \\ \theta \end{array} \right), \quad k=1, 2, \dots$$

① How to select p ? Sample PACF

② Different types of AR models

e.g.: a) Suppose we fit AR(1) to $y_1, \dots, y_n$.

$$\hat{\phi}_0 = 5, \quad \hat{\phi}_1 = \frac{1}{2}$$

SHORT-RANGE
PREDICTIONS

$$\hat{y}_{n+1} = 5 + \frac{1}{2} y_n$$

$$\begin{aligned} \hat{y}_{n+2} &= 5 + \frac{1}{2} \hat{y}_{n+1} = 5 + \frac{1}{2} \left(5 + \frac{1}{2} y_n \right) \\ &= 5 \left(1 + \frac{1}{2} \right) + \frac{1}{4} y_n \end{aligned}$$

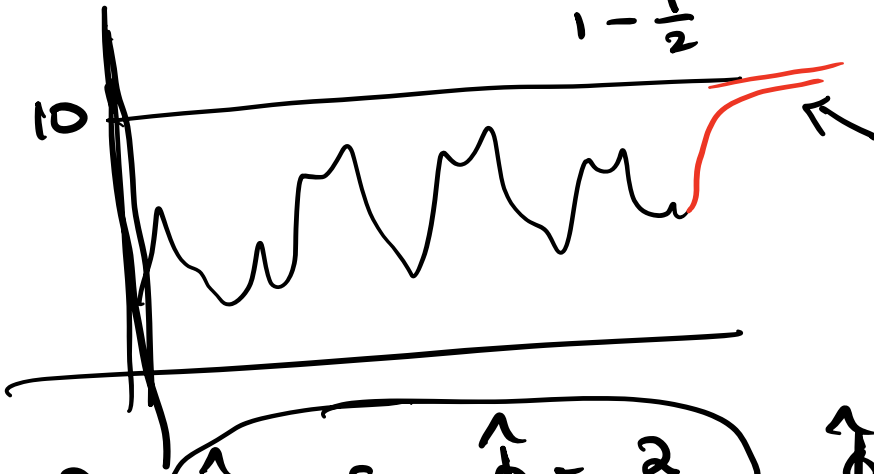


$$\hat{y}_{n+3} = 5 + \frac{1}{2} \hat{y}_{n+2}$$

$$= 5 \left(1 + \frac{1}{2} + \frac{1}{2^2} \right) + \frac{y_n}{2^3}$$

$$\hat{y}_{n+k} = 5 \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{k-1}} \right) + \frac{y_n}{2^k}$$

$$\xrightarrow{k \rightarrow \infty} \frac{5}{1 - \frac{1}{2}} = 10$$



(b) $\hat{\phi}_0 = 5, \hat{\phi}_1 = 2$

$$y_k \rightarrow \infty$$

$$\hat{\phi}_1 = 1.09$$

or
1.1

$$\hat{\phi}_1 = 1.03$$

(c) $\hat{\phi}_0 = 5, \hat{\phi}_1 = 1$

$$\hat{y}_{n+1} = 5 + y_n$$

$$\hat{y}_{n+2} = 5 + y_{n+1} = 10 + y_n$$

$$\hat{y}_{n+3} = 15 + y_n$$

$$\hat{y}_{n+k} = 5k + y_n \rightarrow \text{linear growth in } k$$

AR(1): exploding prediction
settling to a constant
linear predictions

Next Topics

- ① Mathematics of AR(p)
- ② MA (Moving Average)
- ③ ARMA & ARIMA

{ Stationary time series models
Non-stationary models

{ Mean models
Variance models

Definition of Stationary Time Series Models

$$y_t, \quad t = \dots, -3, -2, -1, 0, 1, 2, \dots$$

$(y_1, \dots, y_n)$

Our models in this part of the course are jointly Gaussian.

As a result, the models can be described using means & covariances.

Defn (Stationarity)

$$y_t, \quad t = \dots, -3, -2, -1, 0, 1, 2, \dots$$

is stationary if

(a) $E y_t$ does not change with t

(b) $\text{var } y_t$ does not change with t

(c) $\text{cov}(y_{t_1}, y_{t_2})$ only depends on the gap between t_1 & t_2

ie, $\text{cov}(y_{t_1}, y_{t_2})$ only depends on $|t_1 - t_2|$

$$\text{cov}(y_{99}, y_{100}) = \text{cov}(y_{35}, y_{36})$$

Suppose $\{y_t\}$ is a stationary time series model.

$$\gamma(h) = \text{cov}(y_t, y_{t+h}), \quad h = \dots, -2, -1, 0, 1, 2, \dots$$

↓
ACVF of $\{y_t\}$
Auto Co Variance Function

$$\gamma(0) = \text{cov}(y_t, y_t) = \text{var}(y_t)$$

$$\rho(h) = \text{corr}(y_t, y_{t+h})$$

↑
Auto Correlation
Function
ACF

$$= \frac{\text{cov}(y_t, y_{t+h})}{\sqrt{\text{var}(y_t) \text{var}(y_{t+h})}} = \frac{\gamma(h)}{\sqrt{\gamma(0) \times \gamma(0)}}$$

$$\boxed{\rho(h) = \frac{\gamma(h)}{\gamma(0)}}$$

$$\rho(0) = 1$$

Note: ① **Stationarity** applies to a time series model but not to actual data.

② ACVF & ACF apply only to stationary models

③ Many time series models are not **stationary**

Examples

① $y_t \stackrel{iid}{\sim} N(\mu, \sigma^2)$

$$y_t = \mu + \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$$

STATIONARY

$$\rho(h) = \begin{cases} 1 & \text{if } h=0 \\ 0 & \text{if } h \neq 0 \end{cases}$$

$$\textcircled{2} \quad y_t = \beta_0 + \beta_1 t + \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$$

$$\mathbb{E} y_t = \beta_0 + \beta_1 t \rightarrow \text{changes with } t.$$

$$\textcircled{3} \quad y_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t + \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$$

$$\mathbb{E} y_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t \rightarrow \text{changes with } t$$

$$\textcircled{4} \quad \text{SPECTRUM Model: } \boxed{\text{STATIONARY}}$$

$$y_t = \beta_0 + \sum_{j=1}^m \left[\beta_{j1} \cos \frac{2\pi j t}{n} + \beta_{j2} \sin \frac{2\pi j t}{n} \right]$$

n : sample size

m = largest integer strictly smaller than n .

$$\beta_{j1}, \beta_{j2} \stackrel{iid}{\sim} N(0, \tau_j^2)$$

$$\text{parameters: } \beta_0, \tau_1^2, \dots, \tau_m^2$$

$$\mathbb{E} y_t = \beta_0$$

$$\begin{aligned} & \text{Cov}(y_{t_1}, y_{t_2}) \\ &= \text{Cov} \left(\sum_{j=1}^m \left(\beta_{j1} \cos \frac{2\pi j t_1}{n} + \beta_{j2} \sin \frac{2\pi j t_1}{n} \right), \right. \end{aligned}$$

$$\sum_{j=1}^m \left(\beta_{1j} \cos \frac{2\pi j t_2}{n} + \beta_{2j} \sin \frac{2\pi j t_2}{n} \right)$$

$\beta_{1j}, \beta_{2j}, j=1, \dots, m$ are all independent
 $\beta_{1j}, \beta_{2j} \sim N(0, \tau_j^2)$

$$= \sum_{j=1}^m \text{Cov} \left(\beta_{1j} \cos \frac{2\pi j t_1}{n} + \beta_{2j} \sin \frac{2\pi j t_1}{n}, \beta_{1j} \cos \frac{2\pi j t_2}{n} + \beta_{2j} \sin \frac{2\pi j t_2}{n} \right)$$

$$= \sum_{j=1}^m \tau_j^2 \left\{ \cos \frac{2\pi j t_1}{n} \cos \frac{2\pi j t_2}{n} + \sin \frac{2\pi j t_1}{n} \sin \frac{2\pi j t_2}{n} \right\}$$

$$= \sum_{j=1}^m \tau_j^2 \cos \left(\frac{2\pi j}{n} (t_1 - t_2) \right)$$

$$= \sum_{j=1}^m \tau_j^2 \cos \left(\frac{2\pi j}{n} |t_1 - t_2| \right)$$

$$y_t = \beta_0 + \sum_{j=1}^m \left(\beta_{1j} \cos \frac{2\pi j}{n} t + \beta_{2j} \sin \frac{2\pi j}{n} t \right)$$

$$y_t = \beta_0 + \int_0^{1/2} \left[\beta_1(\omega) \cos 2\pi \omega t + \beta_2(\omega) \sin 2\pi \omega t \right] d\omega$$

$$\beta_1(\omega), \beta_2(\omega) \sim N(0, \sigma^2)$$

Qn: Is AR(1) or AR(p), $p \geq 1$ stationary?

AR(1): $y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t$
 $\varepsilon_t \text{ iid } N(0, \sigma^2)$

Implicit definition

$$x^2 + 5x + 3 = 0$$

$$y_1 = \phi_0 + \phi_1 y_0 + \varepsilon_1$$

$$\begin{aligned} y_2 &= \phi_0 + \phi_1 y_1 + \varepsilon_2 \\ &= \phi_0 + \phi_1 (\phi_0 + \phi_1 y_0 + \varepsilon_1) + \varepsilon_2 \end{aligned}$$

For every $t=1, 2, \dots$
 can write y_t in terms of $y_0, \varepsilon_1, \dots, \varepsilon_t$

$$(y_{-1})$$

$$y_0 = \phi_0 + \phi_1 y_{-1} + \varepsilon_0$$

$$y_{-1} = \frac{y_0 - \phi_0}{\phi_1} - \frac{\varepsilon_0}{\phi_1}$$

$$y_{-2} = \dots$$

Examples: Fix $|\phi_1| < 1$.

$$y_t = \frac{\phi_0}{1 - \phi_1} + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{t-j}$$

→ STATIONARY MODEL

$$y_t = \frac{\phi_0}{1 - \phi_1} + \varepsilon_t + \phi_1 \varepsilon_{t-1} + \phi_1^2 \varepsilon_{t-2} + \phi_1^3 \varepsilon_{t-3} + \dots$$

$$\dots \varepsilon_0, \varepsilon_1, \varepsilon_2, \dots, \varepsilon_{t-2}, \varepsilon_{t-1}, \varepsilon_t$$

$$\downarrow \mathbb{E} y_t = \frac{\phi_0}{1-\phi_1} y_t$$

Check: $\text{Cov}(y_t, y_{t+h}) = \frac{\sigma^2}{1-\phi_1^2} \phi_1^{|h|}$

Check: $y_t = \frac{\phi_0}{1-\phi_1} + \varepsilon_t + \phi_1 \varepsilon_{t-1} + \phi_1^2 \varepsilon_{t-2} + \dots$
 $\varepsilon_t \sim N(0, \sigma^2)$
 satisfies

$$y_t - \phi_1 y_{t-1}$$

$$= \left(\frac{\phi_0}{1-\phi_1} + \varepsilon_t + \phi_1 \varepsilon_{t-1} + \phi_1^2 \varepsilon_{t-2} + \dots \right)$$

$$- \phi_1 \left(\frac{\phi_0}{1-\phi_1} + \varepsilon_{t-1} + \phi_1 \varepsilon_{t-2} + \phi_1^2 \varepsilon_{t-3} + \dots \right)$$

$$= \phi_0 + \varepsilon_t$$

$$y_t = \frac{\phi_0}{1-\phi_1} - \frac{\varepsilon_{t+1}}{\phi_1} - \frac{\varepsilon_{t+2}}{\phi_1^2} - \frac{\varepsilon_{t+3}}{\phi_1^3} - \dots$$

Assume $|\phi_1| > 1$

Check: $y_t - \phi_1 y_{t-1} = \phi_0 + \varepsilon_t$

$\therefore \{y_t\}$ is STATIONARY.

Qn: Is AR(1) stationary?

Ans: $y_t = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{t-j}, |\phi_1| < 1$
→ STATIONARY

$y_t = \frac{\phi_0}{1-\phi_1} - \sum_{j=1}^{\infty} \frac{\varepsilon_{t+j}}{\phi_1^j}, |\phi_1| > 1$
→ STATIONARY

If $|\phi_1| = 1$, then there CANNOT be any stationary AR(1).

e.g.: $\phi_1 = 1$ $y_t = \phi_0 + y_{t-1} + \varepsilon_t$

$y_1 = \phi_0 + y_0 + \varepsilon_1$ →

$y_k = k\phi_0 + y_0 + \underbrace{\varepsilon_1 + \dots + \varepsilon_k}$

SUMMARY

AR(1) → $y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t$

AR(1)
DIFFERENCE
EQUATION

$$x^2 - 4 = 0$$

$|\phi_1| < 1$: UNIQUE STATIONARY SOLUTION

$$y_t = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{t-j}$$

CAUSAL STATIONARY SOLUTION

$$\dots \dots \dots \varepsilon_{t+1}, \varepsilon_t$$

$$\downarrow$$

$$y_t$$

$|\phi_1| > 1$: UNIQUE STATIONARY SOLUTION

$$y_t = \frac{\phi_0}{1 - \phi_1} - \sum_{j \neq 1}^{\infty} \frac{\varepsilon_{t+j}}{\phi_1^j} \quad X$$

→ NON CAUSAL STATIONARY SOLUTION

$|\phi_1| = 1$: NO STATIONARY SOLUTION

$\phi_1 = 1$

Practice: $\phi_1 = 1 \wedge |\phi_1| < 1$